



The study of infinite graphs is an attractive, but often neglected, part of graph theory. This chapter aims to give an introduction that starts gently, but then moves on in several directions to display both the breadth and some of the depth that this field has to offer. Our overall theme will be to highlight the typical kinds of phenomena that will always appear when graphs are infinite, and to show how they can lead to deep and fascinating problems.

Perhaps the most typical such phenomena occur already when the graphs are ‘only just’ infinite, when they have only countably many vertices and perhaps only finitely many edges at each vertex. This is not surprising: after all, some of the most basic structural features of graphs, such as paths, are intrinsically countable. Problems that become really interesting only for uncountable graphs tend to be interesting for reasons that have more to do with sets than with graphs, and are studied in *combinatorial set theory*. This, too, is a fascinating field, but not our topic in this chapter. The problems we shall consider will all be interesting for countable graphs, and set-theoretic problems will not arise.

The terminology we need is exactly the same as for finite graphs, except when we wish to describe an aspect of infinite graphs that has no finite counterpart. One important such aspect is the eventual behaviour of the infinite paths in a graph, which is captured by the notion of *ends*. The ends of a graph can be thought of as additional limit points at infinity to which its infinite paths converge. This convergence is described formally in terms of a natural topology placed on the graph together with its ends. In Sections 6–8 we shall therefore assume familiarity with the basic concepts of point-set topology; reminders of the relevant definitions will be included as they arise.

8.1 Basic notions, facts and techniques

This section gives a gentle introduction to the aspects of infinity most commonly encountered in graph theory.¹

After just a couple of definitions, we begin by looking at a few obvious properties of infinite sets, and how they can be employed in the context of graphs. We then illustrate how to use the three most basic common tools in infinite graph theory: Zorn's lemma, transfinite induction, and something called 'compactness'. We complete the section with the combinatorial definition of an end; topological aspects will be treated in Section 8.6.

A graph is *locally finite* if all its vertices have finite degrees. An infinite graph (V, E) of the form

$$V = \{x_0, x_1, x_2, \dots\} \quad E = \{x_0x_1, x_1x_2, x_2x_3, \dots\}$$

is called a *ray*, and a *double ray* is an infinite graph (V, E) of the form

$$V = \{\dots, x_{-1}, x_0, x_1, \dots\} \quad E = \{\dots, x_{-1}x_0, x_0x_1, x_1x_2, \dots\};$$

in both cases the x_n are assumed to be distinct. Thus, up to isomorphism, there is only one ray and one double ray, the latter being the unique infinite 2-regular connected graph. In the context of infinite graphs, finite paths, rays and double rays are all called *paths*.

The subrays of a ray or double ray are its *tails*. Formally, every ray has infinitely many tails, but any two of them differ only by a finite initial segment. The union of a ray R with infinitely many disjoint finite paths having precisely their first vertex on R is a *comb*; the last vertices of those paths are the *teeth* of this comb, and R is its *spine*. (If such a path is trivial, which we allow, then its unique vertex lies on R and also counts as a tooth; see Figure 8.1.1.)

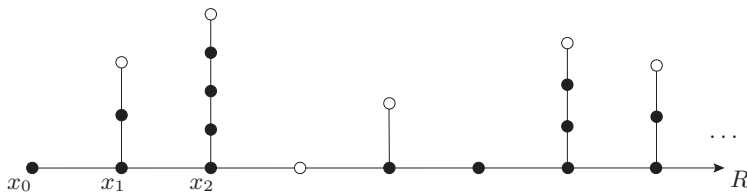


Fig. 8.1.1. A comb with white teeth and spine $R = x_0x_1 \dots$

¹ This introductory section is deliberately kept informal, with the emphasis on ideas rather than definitions that do not belong in a graph theory book. A more formal reminder of those basic definitions about infinite sets and numbers that we shall need is given in Appendix A at the end of the book.

Let us now look at a few very basic properties of infinite sets, and see how they appear in some typical arguments about graphs.

An infinite set minus a finite subset is still infinite. (1)

This trivial property is eminently useful when the infinite set in question plays the role of ‘supplies’ that keep an iterated process going. For example, let us show that if a graph G is infinitely connected (that is, if G is k -connected for every $k \in \mathbb{N}$), then G contains a subdivision of $K^{\aleph_0}$, the complete graph of order $|\mathbb{N}|$. We embed $K^{\aleph_0}$ in G (as a topological minor) in one infinite sequence² of steps, as follows. We begin by enumerating its vertices. Then at each step we embed the next vertex in G , connecting it to the images of its earlier neighbours by paths in G that avoid any other vertices used so far. The point here is that each new path has to avoid only finitely many previously used vertices, which is not a problem since deleting any finite set of vertices keeps G infinitely connected.

$K^{\aleph_0}$

If G , too, is countable, can we then also find a $TK^{\aleph_0}$ as a spanning subgraph of G ? Although embedding $K^{\aleph_0}$ in G topologically as above takes infinitely many steps, it is by no means guaranteed that the $TK^{\aleph_0}$ constructed uses all the vertices of G . However, it is not difficult to ensure this: since we are free to choose the image of each new vertex of $K^{\aleph_0}$, we can choose this as the next unused vertex from some fixed enumeration of $V(G)$. In this way, every vertex of G gets chosen eventually, unless it becomes part of the $TK^{\aleph_0}$ before its time, as a subdividing vertex on one of the paths.

Unions of countably many countable sets are countable. (2)

This fact can be applied in two ways: to show that sets that come to us as countable unions are ‘small’, but also to rewrite a countable set deliberately as a disjoint union of infinitely many infinite subsets. For an example of the latter type of application, let us show that an infinitely edge-connected countable graph has infinitely many edge-disjoint spanning trees. (Note that the converse implication is trivial.) The trick is to construct the trees simultaneously, in one infinite sequence of steps. We first use (2) to partition $\mathbb{N}$ into infinitely many infinite subsets N_i ($i \in \mathbb{N}$). Then at step n we look which N_i contains n , and add a further vertex v to the i th tree T_i . As before, we choose v minimal in some fixed enumeration of $V(G)$ among the vertices not yet in T_i , and join v to T_i by a path avoiding the finitely many edges used so far.

Clearly, a countable set cannot have uncountably many disjoint subsets. However,

² We reserve the term ‘infinite sequence’ for sequences indexed by the set of natural numbers. (In the language of well-orderings: for sequences of order type ω .)

A countable set can have uncountably many subsets whose pairwise intersections are all finite. (3)

This is a remarkable property of countable sets, and a good source of counterexamples to rash conjectures. Can you prove it without looking at Figure 8.1.4?

Another common pitfall in dealing with infinite sets is to assume that the intersection of an infinite nested sequence $A_0 \supseteq A_1 \supseteq \dots$ of infinite (or uncountable) sets must still be infinite (or uncountable). It need not be; in fact it may be empty. (Examples?)

Before we move on to our discussion of common infinite proof techniques, let us look at one more type of construction. One often wants to construct a graph G with a property that is in some sense local, a property that has more to do with the finite subgraphs of G than with G itself. Rather than formalize what exactly this should mean, let us consider an example: given two large integers k and g , let us construct a graph G that is k -connected and has girth at least g .³

We start with a cycle of length g ; call it G_0 . This graph has the right girth, but it is not k -connected. To cure this defect for the vertices of G_0 , join every pair of them by k new independent paths, keeping all these paths internally disjoint. If we choose the paths long enough, the resulting graph G_1 will again have girth g , and no two vertices of G_0 can be separated in it by fewer than k other vertices. Of course, G_1 is not k -connected either. But we can repeat the construction step for the pairs of vertices of G_1 , extending G_1 to G_2 , and so on. The limit graph $G = \bigcup_{n \in \mathbb{N}} G_n$ will again have girth g , since any short cycle would have appeared in some G_n on the way. And, unlike all the G_n , it will be k -connected: since every two vertices are contained in some common G_n , they cannot be separated by fewer than k other vertices in G_{n+1} , let alone in G .

There are a few basic proof techniques that are found frequently in infinite combinatorics. The two most common of these are the use of Zorn's lemma and transfinite induction. Rather than describing these formally,⁴ we illustrate their use by a simple example.

Proposition 8.1.1. *Every connected graph contains a spanning tree.*

First proof (by Zorn's lemma).

Given a connected graph G , consider the set of all trees $T \subseteq G$, ordered by the subgraph relation. Since G is connected, any maximal such tree contains every vertex of G , i.e. is a spanning tree of G .

³ There are finite such graphs, but they are much harder to construct; we shall prove their existence by random methods in Chapter 11.2.

⁴ Appendix A offers brief introductions to both, enough to enable the reader to use these tools with confidence in practice.

To prove that a maximal tree exists, we have to show that for any chain $\mathcal{C}$ of such trees there is an upper bound: a tree $T^* \subseteq G$ containing every tree in $\mathcal{C}$ as a subgraph. We claim that $T^* := \bigcup \mathcal{C}$ is such a tree.

To show that T^* is connected, let $u, v \in T^*$ be two vertices. Then in $\mathcal{C}$ there is a tree T_u containing u and a tree T_v containing v . One of these is a subgraph of the other, say $T_u \subseteq T_v$. Then T_v contains a path from u to v , and this path is also contained in T^* .

To show that T^* is acyclic, suppose it contains a cycle C . Each of the edges of C lies in some tree in $\mathcal{C}$. These trees form a finite subchain of $\mathcal{C}$, which has a maximal element T . Then $C \subseteq T$, a contradiction. $\square$

Transfinite induction and recursion are very similar to finite inductive proofs and constructions, respectively. Basically, one proceeds step by step, and may at each step assume as known what was shown or constructed before. The only difference is that one may ‘start again’ after performing any infinite number of steps. This is formalized by the use of ordinals rather than natural numbers for counting the steps; see Appendix A.

Just as with finite graphs, it is usually more intuitive to construct a desired object (such as a spanning tree) step by step, rather than starting with some unknown ‘maximal’ object and then proving that it has the desired properties. More importantly, a step-by-step construction is almost always the best way to *find* the desired object: only later, when one understands the construction well, can one devise an inductive ordering (one whose chains have upper bounds) in which the desired objects appear as the maximal elements. Thus, although Zorn’s lemma may at times provide an elegant way to wrap up a constructive proof, it cannot in general replace a good understanding of transfinite induction – just as a preference for elegant direct definitions of finite objects cannot, for a thorough understanding, replace the more pedestrian algorithmic approach.

Our second proof of Proposition 8.1.1 illustrates both the constructive and the proof aspect of transfinite induction in the typical intertwined way. We define larger and larger subgraphs $T_\alpha \subseteq G$ inductively. At each step α we prove that T_α is a tree. The definition of T_α will assume and use that subgraphs T_β for all $\beta < \alpha$ have been previously defined, but not only this: it needs to assume that they are nested trees. This fact, therefore, has to be proved along with the recursive definition, always ‘just before’ it is needed.

Second proof (by transfinite induction).

Let G be a connected graph. We define trees $T_\alpha \subseteq G$ recursively so that

$$T_\beta \subseteq T_\alpha \text{ for all } \beta < \alpha. \quad (*_\alpha)$$

Let T_0 consist of a single vertex. Given a limit ordinal $\alpha > 0$, let $T_\alpha := \bigcup_{\beta < \alpha} T_\beta$. Since the T_β are trees satisfying $(*_\beta)$, our new T_α is also a tree (as in the first proof), and it clearly satisfies $(*_\alpha)$.

Given a successor $\alpha = \beta + 1$, we first check whether $G - T_\beta = \emptyset$. If so, then T_β is a spanning tree and we terminate the recursion. If not, then $G - T_\beta$ has a vertex v_α that sends an edge e_α to a vertex in T_β . Then T_α , obtained from T_β by adding v_α and e_α , is a tree satisfying $(*_\alpha)$.

It remains to check that our recursion does indeed terminate. But if $v_{\beta+1}$ gets defined for all $\beta < \gamma$ then $\beta \mapsto v_{\beta+1}$ is an injective map showing that $|\gamma| \leq |G|$. This cannot hold for all ordinals γ ; it fails, for example, when γ represents a well-ordering of the power set of $V(G)$. $\square$

Why did these proofs work so smoothly? The reason is that the forbidden substructures, cycles, were finite and therefore could not arise unexpectedly at limit steps. If we wanted to construct a *rayless* spanning tree, on the other hand, one that contains no ray, then the edges of partial finite trees T_β might combine to form a ray in $T_\alpha = \bigcup_{\beta < \alpha} T_\beta$ when α is a limit. And indeed, here lies the challenge in most transfinite constructions: to make the right choices at successor steps to ensure that the structure will also be as desired at limits.

*compactness
proofs*

Our third basic proof technique, somewhat mysteriously referred to as *compactness* (see below for why), offers a formalized way of making the right choices in certain standard cases. These are cases where nothing unexpected happens at limits, but a choice that looks good at the time it is made may lead to a dead end after another *finite* number of steps – unlike the creation of a cycle, which is visible at once.

For example, let G be a graph whose finite subgraphs are all k -colourable. It is natural then to try to construct a k -colouring of G as a limit of k -colourings of its finite subgraphs. Now each finite subgraph will have several k -colourings; will it matter which we choose? Clearly, it will. When $G' \subseteq G''$ are two finite subgraphs and u, v are vertices of G' that receive the same colour in every k -colouring of G'' (and hence also in any k -colouring of G), we must not give them different colours in the colouring we choose for G' , even if such a colouring exists. However if we do manage, somehow, to colour the finite subgraphs of G compatibly, we shall automatically have a colouring of all of G .

All compactness proofs deal with situations similar to this. We wish to solve a problem about an infinite structure, and we know how to solve it for all the finite substructures. ‘Compactness’ enables us to combine these partial solutions to an overall solution if the partial solutions are compatible in the right way.

For countable structures, all this – the choices, the dead ends, the compatibility requirement on the finite solutions, and how they combine to an overall solution – can be made visible in a particularly intuitive way, by a graph:

Lemma 8.1.2. (König's Infinity Lemma)

Let $V_0, V_1, \dots$ be an infinite sequence of disjoint non-empty finite sets, and let G be a graph on their union. Assume that every vertex v in a set V_n with $n \geq 1$ has a neighbour $f(v)$ in V_{n-1} . Then G contains a ray $v_0 v_1 \dots$ with $v_n \in V_n$ for all n .

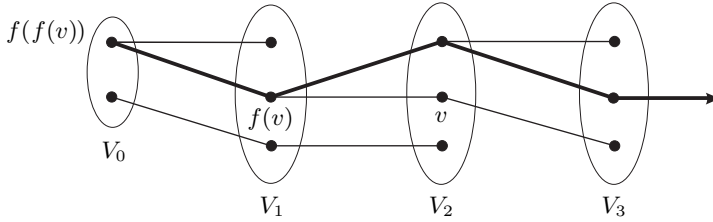


Fig. 8.1.2. König's infinity lemma

Proof. Let $\mathcal{P}$ be the set of all finite paths of the form $v f(v) f(f(v)) \dots$ ending in V_0 . Since V_0 is finite but $\mathcal{P}$ is infinite, infinitely many of the paths in $\mathcal{P}$ end at the same vertex $v_0 \in V_0$. Of these paths, infinitely many also agree on their penultimate vertex $v_1 \in V_1$, because V_1 is finite. Of those paths, infinitely many agree even on their vertex v_2 in V_2 – and so on. Although the set of paths considered decreases from step to step, it is still infinite after any finite number of steps, so v_n gets defined for every $n \in \mathbb{N}$. By definition, each vertex v_n is adjacent to v_{n-1} on one of those paths, so $v_0 v_1 \dots$ is indeed a ray. $\square$

The following ‘compactness theorem’, the first of its kind in graph theory, answers our question about colourings:

Theorem 8.1.3. (de Bruijn & Erdős, 1951)

Let $G = (V, E)$ be a graph and $k \in \mathbb{N}$. If every finite subgraph of G has chromatic number at most k , then so does G .

First proof (for G countable, by the infinity lemma).

Let $v_0, v_1, \dots$ be an enumeration of V and put $G_n := G[v_0, \dots, v_n]$. Write V_n for the set of all k -colourings of G_n with colours in $\{1, \dots, k\}$. Define a graph on $\bigcup_{n \in \mathbb{N}} V_n$ by inserting all edges cc' such that $c \in V_n$ and $c' \in V_{n-1}$ is the restriction of c to $\{v_0, \dots, v_{n-1}\}$. Let $c_0 c_1 \dots$ be a ray in this graph with $c_n \in V_n$ for all n . Then $c := \bigcup_{n \in \mathbb{N}} c_n$ is a colouring of G with colours in $\{1, \dots, k\}$. $\square$

Applications of the infinity lemma such as this one rely on the fact that a countable graph can be exhausted by a nested sequence of finite subgraphs. Appendix A offers a version of the infinity lemma that works for arbitrary graphs, in which these finite subgraphs need not be sequentially ordered. This general version is still very intuitive and can be used conveniently in many settings, including a proof of Theorem 8.1.3.

The essence of compactness proofs is often encoded more directly, if less graphically, in terms of just sets and functions. The *compactness principle* from Appendix A is such a version that is particularly easy to apply. We illustrate its application by another proof of Theorem 8.1.3:

Second proof (for arbitrary graphs, by the compactness principle).

Let $X := V$ and $S := \{1, \dots, k\}$. Let $\mathcal{F}$ be the set of all finite subsets of V . For each $Y \in \mathcal{F}$ let $\mathcal{A}(Y)$ be the set of k -colourings of $G[Y]$. Our proof of Theorem 8.1.3 will be complete once we have found a function $V \rightarrow \{1, \dots, k\}$ that induces a k -colouring on every finite subgraph $G[Y]$, as any such function is clearly a k -colouring of G .

By the compactness principle it suffices to show that, given any finite $\mathcal{Y} \subseteq \mathcal{F}$, we can find a function $V \rightarrow \{1, \dots, k\}$ that induces a colouring on every $G[Y]$ with $Y \in \mathcal{Y}$. But this is easy: just take a k -colouring of the finite graph $G[\bigcup \mathcal{Y}]$, and extend it arbitrarily to the rest of V . $\square$

Our last proof of Theorem 8.1.3 appeals directly to compactness as defined in topology. Recall that a topological space is *compact* if its closed sets have the ‘finite intersection property’, which means that the overall intersection $\bigcap \mathcal{A}$ of a set $\mathcal{A}$ of closed sets is non-empty whenever every finite subset of $\mathcal{A}$ has a non-empty intersection. By Tychonoff’s theorem of general topology, any product of compact spaces is compact in the usual product topology.

Third proof (for arbitrary graphs, by Tychonoff’s theorem).

Consider the product space

$$X := \prod_V \{1, \dots, k\} = \{1, \dots, k\}^V$$

of $|V|$ copies of the finite set $\{1, \dots, k\}$ endowed with the discrete topology. By Tychonoff’s theorem, this is a compact space. Its basic open sets have the form

$$O_h := \{f \in X : f|_U = h\},$$

where h is some map from a finite set $U \subseteq V$ to $\{1, \dots, k\}$.

For every finite set $U \subseteq V$, let A_U be the set of all $f \in X$ whose restriction to U is a k -colouring of $G[U]$. These sets A_U are closed (as well as open – why?), and for any finite set $\mathcal{U}$ of finite subsets of V we have $\bigcap_{U \in \mathcal{U}} A_U \neq \emptyset$, because $G[\bigcup \mathcal{U}]$ has a k -colouring. By the finite intersection property of the sets A_U , their overall intersection is non-empty, and every element of this intersection is a k -colouring of G . $\square$

Although our three compactness proofs look formally different, it is instructive to compare them in detail, checking how the requirements in one are reflected in the other (cf. Exercise 18).

As mentioned before, the standard use for compactness proofs is to transfer theorems from finite to infinite graphs, or conversely. This is not always quite as straightforward as above; often, the statement has to be modified a little to make it susceptible to a compactness argument.

As an example – see Exercises 19–30 for more – let us prove the locally finite version of the following famous conjecture. Call a bipartition of the vertex set of a graph *unfriendly* if every vertex has at least as many neighbours in the other class as in its own. Clearly, every finite graph has an unfriendly partition: just take any partition that maximizes the number of edges between the partition classes. At the other extreme, it can be shown by set-theoretic methods that uncountable graphs need not have such partitions. Thus, intriguingly, it is the countable case that has remained unsolved:

Unfriendly Partition Conjecture. *Every countable graph admits an unfriendly partition of its vertex set.*

Proof for locally finite graphs. Let $G = (V, E)$ be an infinite but locally finite graph, and enumerate its vertices as $v_0, v_1, \dots$. For every $n \in \mathbb{N}$, let $\mathcal{V}_n$ be the set of partitions of $V_n := \{v_0, \dots, v_n\}$ into two sets U_n and W_n such that every vertex $v \in V_n$ with $N_G(v) \subseteq V_n$ has at least as many neighbours in the other class as in its own. Since the conjecture holds for finite graphs, the sets $\mathcal{V}_n$ are non-empty. For all $n \geq 1$, every $(U_n, W_n) \in \mathcal{V}_n$ induces a partition (U_{n-1}, W_{n-1}) of V_{n-1} , which lies in $\mathcal{V}_{n-1}$. By the infinity lemma, there is an infinite sequence of partitions $(U_n, W_n) \in \mathcal{V}_n$, one for every $n \in \mathbb{N}$, such that each is induced by the next. Then $(\bigcup_{n \in \mathbb{N}} U_n, \bigcup_{n \in \mathbb{N}} W_n)$ is an unfriendly partition of G . $\square$

The trick that made this proof possible was to require, for the partitions of V_n , correct positions only of vertices that send no edge out of V_n : this weakening is necessary to ensure that partitions from $\mathcal{V}_n$ induce partitions in $\mathcal{V}_{n-1}$; but since, by local finiteness, every vertex has this property eventually (for large enough n), the weaker assumption suffices to ensure that the limit partition is unfriendly.

Let us complete this section with an introduction to the one important concept of infinite graph theory that has no finite counterpart, the notion of an end. An *end*⁵ of a graph G is an equivalence class of rays in G , where two rays are considered equivalent if, for every finite set $S \subseteq V(G)$, both have a tail in the same component of $G - S$. This

end

⁵ Not to be confused with the ends, or endvertices, of an edge. In the context of infinite graphs, we use the term ‘endvertices’ to avoid confusion.